FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
CONCEPT RECAPITULATION TEST – III
PAPER –1
TEST DATE: 26-04-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. BC

Sol. $F = qE = -q\dfrac{dv}{dx}$

$\dfrac{1}{2}mv_0^2 = q\left(10^4\right)$

2. ABD

Sol. $E_{eff} = \dfrac{\dfrac{E_1}{r_1} + \dfrac{E_2}{r_2}}{\dfrac{1}{r_1} + \dfrac{1}{r_2}} = 20V$

$R_{eff} = 2\Omega$

3. BCD

Sol. $J.\dfrac{a}{\sqrt{2}} = I\omega$

$v_A = v_{cm} - \omega.\dfrac{a}{\sqrt{2}}$

$v_B = v_{cm} + \omega.\dfrac{a}{\sqrt{2}}$

4. ABCD

Sol. $mv_1 = 2mv_2$

$\dfrac{1}{2}mv_1^2 + \dfrac{1}{2}\cdot 2m\cdot v_2^2 = mgR$

$v_1 = 2\sqrt{\dfrac{gR}{3}},\ v_2 = \sqrt{\dfrac{gR}{3}}$

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5. AC

Sol. in free expansion, temperature is constant.

$\Rightarrow P_1V_1 = P_2V_2$

6. A

Sol. $R_{eff} = 3R$

$$Q_{max} = \frac{CV}{2}$$

$$q = Q_{max}\left(1 - e^{-\frac{t}{R_{eff}.C}}\right)$$

7. A

Sol. $$I_X = \frac{5}{4}MR^2 + \frac{5}{4}MR^2 + \frac{MR^2}{4} + M\left(R + R\sqrt{3}\right)^2$$

$$= \frac{11MR^2}{4} + MR^2 + 3MR^2 + 2MR^2\sqrt{3}$$

$$= \frac{\left(27 + 8\sqrt{3}\right)MR^2}{4}$$

$$I_Y = \frac{MR^2}{4} + \frac{5}{4}MR^2 + \frac{5}{4}MR^2 = \frac{11MR^2}{4}$$

$$I_C = 3\left\{\frac{MR^2}{2} + M\left(\frac{2R}{\sqrt{3}}\right)^2\right\} = \frac{3MR^2}{2} + 4MR^2 = \frac{11MR^2}{2}$$

$$I_A = \frac{11MR^2}{2} + 3M\left(R + \frac{2R}{\sqrt{3}}\right)^2 = \frac{11MR^2}{2} + 3MR^2 + 4MR^2 + 4\sqrt{3}MR^2$$

$$= \frac{\left(25 + 4\sqrt{3}\right)}{2}MR^2$$

8. C

Sol. Basic concept.

$p = Q.d$

9. C

Sol. Basic concept of symmetry and combination of capacitors.

10. A

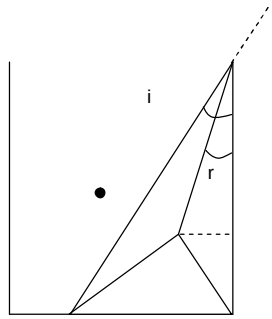
Sol. Basic concept.

Section – B

11. 1.92

Sol. $\tan i = \frac{3a}{9a} = \frac{1}{3}$

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$\tan r = \frac{a}{6a} = \frac{1}{6}$

$\frac{\sin i}{\sin r} = \mu$

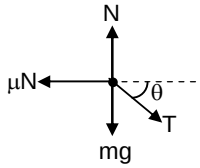
12. 0.50

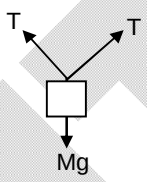
Sol. $2T\sin\theta = Mg$

$\mu N \geq T\cos\theta$

$N = mg + T\sin\theta$

$\cot\theta \leq \mu\left(\frac{2m+M}{M}\right)$





13. 25.00

Sol. $v' = 2\times 5 + 10 = 20;\ h = \frac{v^2}{2g} = \frac{20\times 20}{20} = 20,\quad H = 20 + 5 = 25$

14. 3.70

Sol. $M\sqrt{2g\ell} + O = M\cdot v_1 + m.\sqrt{5g\ell}$

$\sqrt{2g\ell} = \sqrt{5g\ell} - v$

$\frac{M}{m} = \frac{\sqrt{5}}{2\sqrt{2}-\sqrt{5}}$

15. 0.75

Sol. $F\cdot R = f\,R$

$F + F\cos 57^0 = mg\sin 37^0$

$\frac{8F}{5} = mg\times\frac{3}{5} \Rightarrow F = \frac{3mg}{8}$

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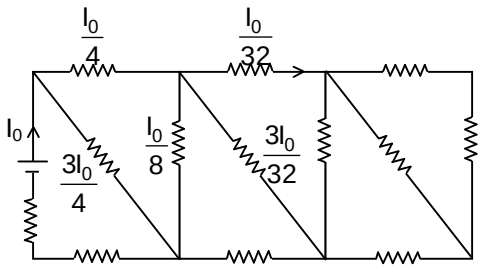
16. 79.30

Sol. $E = 10 \times 7 + \frac{1}{100} \times 930 = 79.3$

17. 32.00

Sol. $R_{eq} = 6\Omega + r$

$$I_0 = \frac{E}{6 + r}$$



18. 0.74

Sol. $\Delta E = \Delta M \cdot C^2$

$\Delta M = (236.045562) - (232.038054 + 4.002603)$

$= 0.004905$

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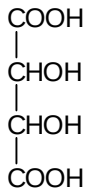
Chemistry

PART – II

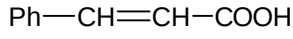
Section – A

19. A

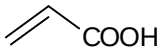
Sol.

(A) 

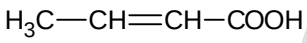
Tartaric acid

(B) 

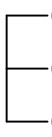
3-phenyl prop-2-enoic acid

(C) 

Acrylic acid



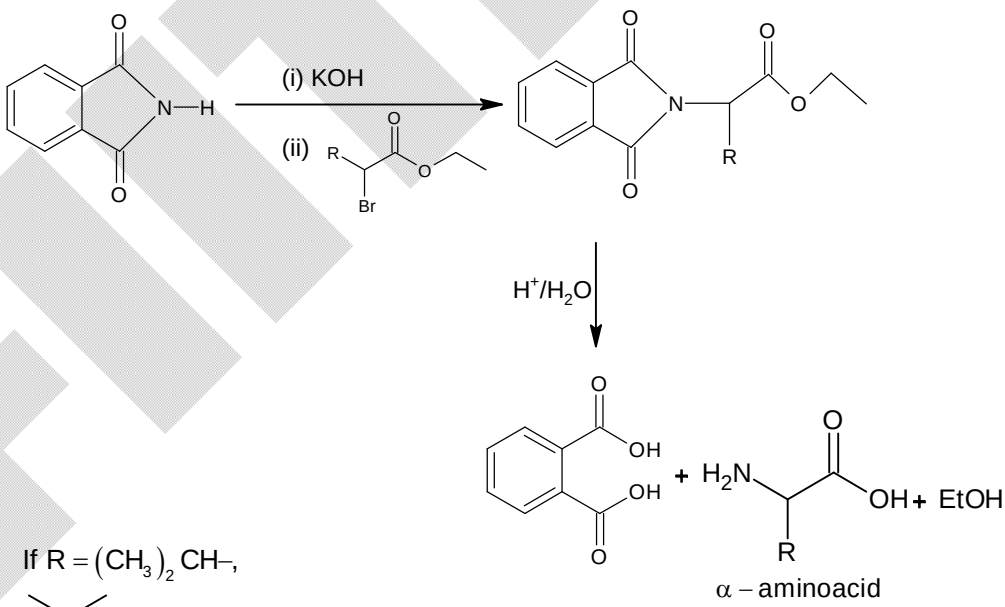
Crotonic acid

(D) 

Propane-1,2,3-Tricarbonitrile

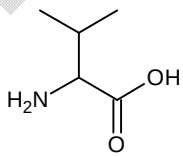
20. AD

Sol.

(A) 

α – aminoacid

(D) If $R = (CH_3)_2CH–$,

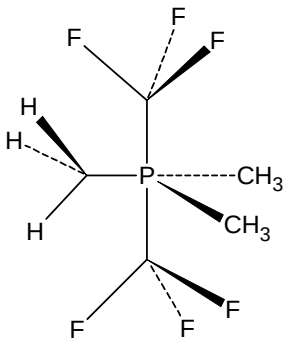


valine

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21. ABCD

Sol.



22. AB

Sol.

$$A_2 + B_2 \rightleftharpoons 2AB$$

t = 0	0.6	0.6	
t = eq	0.6 – x	0.6 – x	2x

$$K_p = \frac{(2x)^2}{(0.6-x)^2} = 64$$

$$\frac{2x}{0.6-x} = 8$$

$$2x = 4.8 - 8x$$

$$x = 0.48$$

$$p_{A_2} = 0.12\,atm$$

$$p_{B_2} = 0.12\,atm$$

$$p_{AB} = 0.96\,atm$$

23. BCD

Sol. Theory based

24. ACD

Sol. $CuCO_3$ is green, rest are white.

25. D

Sol. Theory based

26. C

Sol. Wurtz reaction is best for preparation of symmetrical alkane. Corey – house synthesis can be used to prepare both symmetrical and unsymmetrical alkane.

27. D

Sol.

	SM	**μ**
Mg^{2+}	0	0
Zn^{2+}	0	0
Mn^{2+}	6	$\sqrt{35}$ B.M
Fe^{2+}	5	$\sqrt{24}$ B.M

28. C

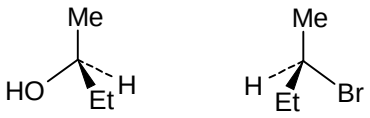
Sol. Theory based

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Section – B

29. 53.77

Sol.



S R

$$\frac{92.46}{2} + 7.54 = 53.77$$

30. 0.04

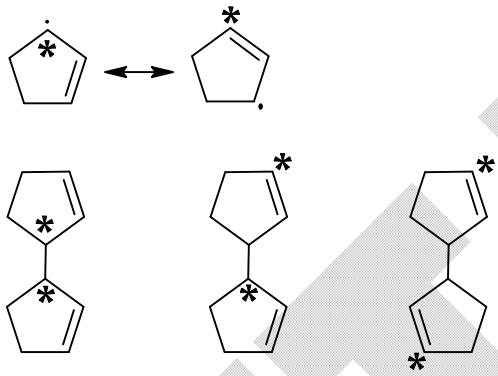
Sol. x = Urotropine

$(CH_2)_6 N_4$

$$\frac{x}{100} = \frac{4}{100} = 0.04$$

31. 3.00

Sol.



32. 110.00

Sol. $\Delta H_r = (1\times 620 + 4\times 410 + 1\times 440) - (1\times 350 + 6\times 410) = -110 \text{ kJ/mole}$

33. 9.00

Sol. $E^0_{cell} = -0.8 + 1.39 = 0.59V$

$$0.59 = \frac{0.059}{1}\log_{10} K$$

$\log_{10} K = 10$

$K = 1\times 10^{10}$

$(y - x) = 10 - 1 = 9$

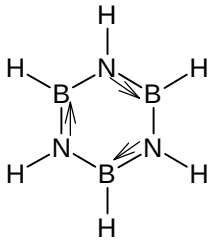
34. 2.00

Sol. $Xe_{(g)} + F_{2(g)} \xrightarrow[1 \text{ bar}]{673 \text{ K}} XeF_{2(g)}$

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35. 3.00

Sol.



36. 90.72

Sol. $$= \frac{\left(6 \times \frac{1}{3} + 1\right)\pi r^2}{6 \times \frac{\sqrt{3}}{4}(2r)^2}$$

$$= 0.9072$$

or 90.72%

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Mathematics

PART – III

Section – A

37. ABC

Sol. $a_1 + a_4 + a_7 + ... + a_{16} = 147$

$\Rightarrow \quad 2a_1 + 15d = 49$

$\Rightarrow \quad a_1 + a_2 + a_3 + a_4 + ... + a_{16} = 392$

$\Rightarrow \quad N = 98;\ M = 49$

Using $A.M \geq G.M \Rightarrow \frac{\Sigma a_i}{16} \geq (a_1 \cdot a_2 .....a_{16})^{\frac{1}{16}} \Rightarrow a_1 \cdot a_2 .....a_{16} \leq \left(\frac{49}{2}\right)^{16}$

$\therefore$ S = 49 and W = 2

38. ABD

Sol. $O(p,q); H(\alpha+4, \beta-3) \text{ and } G\left(\frac{3+\alpha}{3}, \frac{\beta-8}{3}\right)$

using ratio $HG : GO = 2:1$ we get $p = -\frac{1}{2}$ and $q = -\frac{5}{2}$

$m_{AH} = \frac{\beta-2}{\alpha-1}$ and $m_{BC} = \frac{\beta+7}{-2-\alpha} \Rightarrow \beta^2 - \alpha^2 + 5\beta - \alpha = 12$

39. ACD

Sol. Let $f(x) = \frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8}$ and $f(x) + \frac{a}{30} = 0$ ....(i)

$\therefore \quad f(x) = \frac{(x-3)}{(x+4)}$ wher $x \neq 1, 2, , -4$

Range of $f(x) = R - \left\{1, -\frac{2}{5}, -\frac{1}{6}\right\}$

So, equation (i) does not have solution if $\frac{a}{30} = -1, \frac{2}{5}, \frac{1}{6} \Rightarrow a = -30, 12, 5$

40. ACD

Sol. Let degree of f(x) be n therefore $n^2 = n + 2 \Rightarrow n = 2$

Now let $f(x) = ax^2 + bx$

On comparing we get $f(x) = \frac{1}{\sqrt{3}}x^2$

41. BC

Sol. For QA + QB to be minimum, using triangle inequality $QA + QB \geq AB$.

$(QA + QB)_{min} = AB$, now image of point B in the given plane $B'\left(\frac{7}{3}, -\frac{1}{3}, \frac{7}{3}\right)$

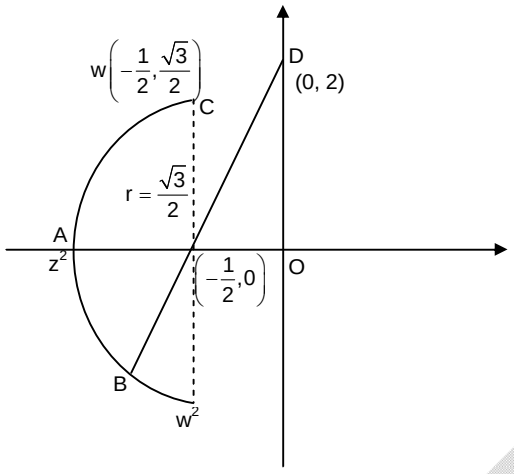
Now, solving plane with line AB' we get $Q\left(\frac{7}{5}, -\frac{1}{5}, \frac{7}{5}\right) \therefore a - c - 10b = 2$

For $|QA - QB|$ is maximum, using triangle inequality we get $|QA - QB|_{max} = AB$ therefore points A, B and Q are collinear, now solving equation of line AB with plane we get $Q(3,3,3) \therefore a - 2b + c = 0$.

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42. BC

Sol. From given relation $\arg\left(\frac{z^2-w}{z^2-w^2}\right)=\frac{\pi}{2}\Rightarrow$ locus of z is semi-circle whose diameter ends are $w=-\frac{1}{2}+\frac{\sqrt{3}}{2}i$ and $W^2=-\frac{1}{2}-\frac{\sqrt{3}}{2}i\Rightarrow\therefore$ centre is $\left(-\frac{1}{2},0\right)$ and radius $=\frac{\sqrt{3}}{2}$



$\therefore\quad |z^2|_{max}=OA=\frac{1}{2}+\frac{\sqrt{3}}{2}\Rightarrow |z|_{max}=\sqrt{\frac{1+\sqrt{3}}{2}}=\frac{\sqrt{2+2\sqrt{3}}}{2}$

$\because\quad |z-1-i||z+1+i|=|(z-1-i)(z+1+i)|=|z^2-(1+i)^2|$

$\Rightarrow\quad |z^2-2i|_{max}=BD=\frac{\sqrt{3}}{2}+\sqrt{\frac{1}{4}+4}=\frac{\sqrt{17}+\sqrt{3}}{2}$

43. B

Sol. (I) Let $z = x + iy\ \therefore\ Im(Z^2)=3\Rightarrow 2xy=3$ which is a rectangular hyperbola of eccentricity $\sqrt{2}$.

(II) Let ends of latus rectum are $\left(ae,\pm\frac{b^2}{a}\right)$ and other focus is $(-ae,0)$ then $\tan 30^0=\frac{b^2/a}{2ae}$

$\Rightarrow 2a^2e=b^2\sqrt{3}$ and $b^2=a^2\left(e^2-1\right)\therefore 2e=\sqrt{3}e^2-\sqrt{3}\Rightarrow e=\sqrt{3}$

(III) AB = 6 = 2ae, PA – PB = 4 = 2a $\therefore$ eccentricity of hyperbola is $e=\frac{3}{2}$

Let eccentricity of conjugate hyperbola is e' then $\frac{1}{e^2}+\frac{1}{(e')^2}=1\Rightarrow e'=\frac{3}{\sqrt{5}}$

(IV) Angle between asymptotes $=\tan^{-1}\left|\frac{2ab}{a^2-b^2}\right|=\frac{\pi}{3}\Rightarrow\left|\frac{\frac{2a}{b}}{\frac{a^2}{b^2}-1}\right|=\sqrt{3}$

Let eccentricity of conjugate hyperbola is e' and $a^2=b^2\left((e')^2-1\right)$

$\therefore\quad \frac{2\sqrt{(e')^2-1}}{|(e')^2-2|}=\sqrt{3}\Rightarrow e'=2$

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44. B

Sol. Domain of f(x) is [0, 1]

and $f(x)=\begin{cases}\dfrac{\pi}{4}, & x\in\left[0,\dfrac{1}{2}\right]\\ 2\sin^{-1}(2x-1)+\dfrac{\pi}{4}, & x\in\left(\dfrac{1}{2},1\right)\\ \pi+\tan^{-1}\left(\dfrac{1}{2}\right), & x=1\end{cases}$

Now verify the options.

45. C

Sol. (I) Let A(a, 0, 0), B(0, b, 0) and C(0, 0, c) then a = 1, b = 2, c = 3 and equation of plane is $\frac{x}{1}+\frac{y}{2}+\frac{z}{3}=1$

(II) Let the plane is $(x + y + 2z - 7) + \lambda (2x - y + z - 6) = 0$ and $(1 + 2\lambda). 2 + (1 - \lambda). 3 + (2 + \lambda). 1 = 0$

$\therefore$ plane is $12x - 99y + 3z - 28 = 0$

(III) Required plane is $\begin{vmatrix} x-3 & y+1 & z+2\\ 2 & -3 & 1\\ -3 & 1 & 2\end{vmatrix}=0\Rightarrow x+y+z=0$

(IV) Let plane is $a(x-1)+b(y+1)+c(z-2)=0$ since this is parallel to given lines $2a+3b+4c=0, a+2b+c=0$ now on eliminating a, b, c we get $5x-2y-z=5$.

46. D

Sol. (I) $6x-x^2+10>3\Rightarrow (x+1)(x-7)<0\Rightarrow x=0,1,2,3,4,5,6$.

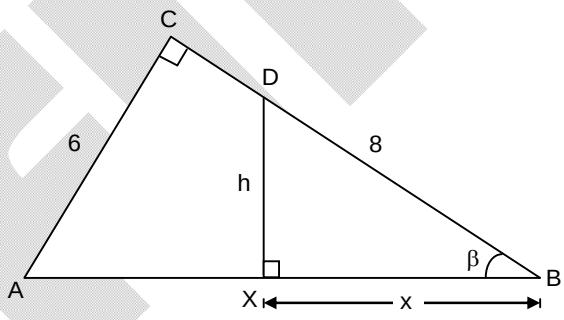
(II). $f(x)=(x-1)(x^2-7x+13)$ for it to be prime, at least of the factors must be prime

$\therefore x-1=1$ or $x^2-7x+13=1\Rightarrow x=2,3,4$

(III). For integer roots D must be perfect square of an integer.

$D=a^2-4(a+1)=l^2\Rightarrow a=5,\ p=-1$ and $a=-1,\ p=1$

(IV). Here $\angle C=90^0$. Let height be h and $\therefore h=x\tan B$



Area of $\Delta BXD=\frac{1}{2}hx\Rightarrow 2.\frac{1}{2}hx=\frac{1}{2}6.8\Rightarrow x^2=24\cot B=\frac{24.8}{6}=32\Rightarrow x=4\sqrt{2}$

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Section – B

47. 14.00

Sol. Solving H and L we get $2x^2+2x-1=0$ let the roots be α and β

$\Rightarrow \alpha+\beta=-1$ and $\alpha.\beta=-\frac{1}{2}$

Let points of intersection are $A(\alpha,-1-\alpha)$ and $B(\beta,-1-\beta)$

Family of circles passing through A and B is $x^2+y^2+x(1+\lambda)+y(1+\lambda)+\lambda-1=0$

$\because$ this circle touches at $A(\alpha,-1-\alpha)$

$\therefore \left(\frac{dy}{dx}\right)_A \cdot m_{AC}=-1$, where $\left(\frac{dy}{dx}\right)_A$ is slope of hyperbola at A and m_{AC} is slope of line joining A and centre of circle C. $\left(\frac{dy}{dx}\right)_A=\frac{\alpha-2}{\alpha+3}$ and $m_{AC}=\frac{\lambda-1-2\alpha}{\lambda+1+2\alpha}$

$\left(\frac{dy}{dx}\right)_A \cdot m_{AC}=-1 \Rightarrow \frac{\alpha-2}{\alpha+3}\times\frac{\lambda-1-2\alpha}{\lambda+1+2\alpha}=-1$

$\Rightarrow \lambda(2\alpha+1)=-5-10\alpha$

$\Rightarrow \lambda=-5$

$\therefore$ circle: $x^2+y^2-4x-4y-6=0$

$\Rightarrow R^2=14$

48. 190.00

Sol. $\because B^2=I$ and $AB=\begin{bmatrix} a_{13} & a_{12} & a_{11} \\ a_{23} & a_{22} & a_{21} \\ a_{33} & a_{32} & a_{31} \end{bmatrix}$ and $AB=AB^3=AB^5=\cdots,+AB^{19}$

$\text{trace}\left(AB+AB^3+AB^5+\cdots\ldots+AB^{19}\right)=210$

$\Rightarrow 10(a_{31}+a_{22}+a_{13})=210$

$\therefore a_{31}+a_{22}+a_{13}=21$

$\therefore$ number of ordered triplets $(a_{31},a_{22},a_{13})=190$

49. 13.00

Sol. $x^6\left(x^3-6x^2+12x-8\right)=0 \Rightarrow x=0,0,0,0,0,0,2,2,2$

$\therefore |A|=\begin{vmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{vmatrix}=8$ and $|B|=6^3$ $\therefore \frac{|\text{adj } B|}{|\text{adj}(\text{adj}(\text{adj } A))|}=\frac{|B|^2}{|A|^{2^3}}=\frac{6^6}{8^8}=\left(\frac{3}{8}\right)^6$

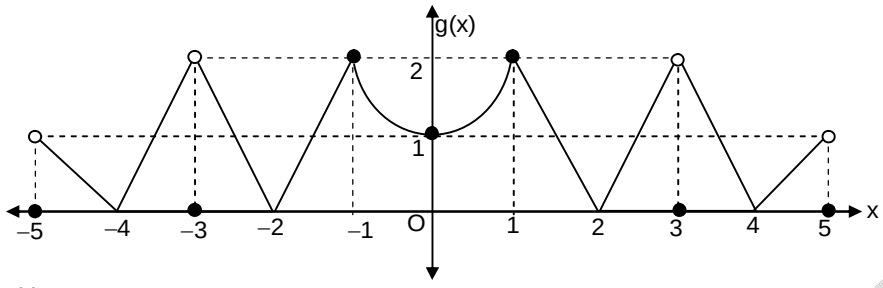
$\Rightarrow p=3$ and $q=8$

50. 16.00

Sol. $\text{sgn}\left(x^2-(k+1)x+(k^2+1)\right)=1$

$\therefore g(g(x))=1$ now let $g(x)=t$

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$\therefore g(t)=1 \Rightarrow t=0, \alpha_1,\alpha_2,\alpha_3,\alpha_4,\alpha_5,\alpha_6$

where $1<\alpha_1<2;\ 2<\alpha_2<3;3<\alpha_3<4$ and $\alpha_4,\alpha_5,\alpha_6<0$

g(x) = 0 has 8 solutions

$g(x)=\alpha_1$ has 8 solutions

Hence total number of solutions is 16.

51. 2.50

Sol. $\int_0^1 f(x)dx=1$

$\Rightarrow \int_0^1 f(2x)dx=3$

Let $2x=t$

$\Rightarrow 2dx=dt$

$\Rightarrow \int_0^2 f(t)dt=6$

$\Rightarrow \int_0^1 f(t)dt+\int_1^2 f(t)dt=6$

$\therefore \frac{1}{2}\int_1^2 f(x)dx=\frac{5}{2}$

52. 720.00

Sol. Let the roots be $a-ib, a+ib$ then $a^2+b^2=1$

$\because$ $H.M=2 \Rightarrow \frac{1}{a-ib}+\frac{1}{a+ib}=\frac{2}{2} \Rightarrow 2a=a^2+b^2$

From two equations we get $a=\frac{1}{2}$

Required quadratic equation is $x^2-2ax+a^2+b^2=0 \Rightarrow x^2-x+1=0$

Given $x^2-\left(\log_2\frac{\alpha}{\beta}\right)x+\cos\alpha-\sin\beta=0$. Now comparing these two equations

$\log_2\frac{\alpha}{\beta}=1$ and $\cos\alpha-\sin\beta=1 \Rightarrow \alpha=2\beta$ and $\cos 2\beta-\sin\beta=1$

$\Rightarrow 2\sin^2\beta+\sin\beta=0 \quad \Rightarrow \sin\beta=-\frac{1}{2}$ or $\sin\beta=0$

$\Rightarrow \beta=210^0,330^0,180^0$

$\therefore$ Sum = 720

53. 19.00

Sol. $f(1)=5, f(2)=8, f'(1)=3$ and $f''(1)=0$

$\therefore f(x)=(x-1)^3(x-2)+3x+2$

$\Rightarrow f(3)=19$

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54. 2024.00

Sol. Let $I=\int_0^1\left(x-f(x)\right)^{2022}dx$ …(i)

let $x=f(t)\Rightarrow dx=f'(t)dt$

$I=\int_1^0\left(f(t)-f\left(f(t)\right)\right)^{2022}f'(t)dt$

$\because f\left(f(0)\right)=0\Rightarrow f(1)=0$

$I=-\int_0^1\left(f(t)-t\right)^{2022}f'(t)dt$ ...(ii)

Now adding equation (1) and (2)

$2I=\int_0^1\left(x-f(x)\right)^{2022}\left(1-f'(x)\right)dx$

$\Rightarrow I=\frac{1}{2023}=\frac{p}{q}$

$\therefore p+q=2024$

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